

AWS Cloud Support Troubleshooting Project

Hands-on VPC, ALB, Auto Scaling, S3, IAM, CloudWatch & SNS Lab

Project focus

- Cloud Support troubleshooting and infrastructure operations
- Real configuration, testing, incident simulation and cleanup
- Evidence captured from AWS Console and Linux CLI

Scope note

- Integrated lab intentionally omitted RDS, NAT Gateway and public Route 53 mapping
- Those items are not presented as completed

1. Project Objective

Build and troubleshoot a small, support-oriented AWS web environment.

What was built

- Custom VPC with public and private subnet structure
- Application Load Balancer with multi-AZ subnet mapping
- EC2 web servers and Auto Scaling Group
- S3 access from EC2 using an IAM role
- CloudWatch CPU monitoring with SNS email notification

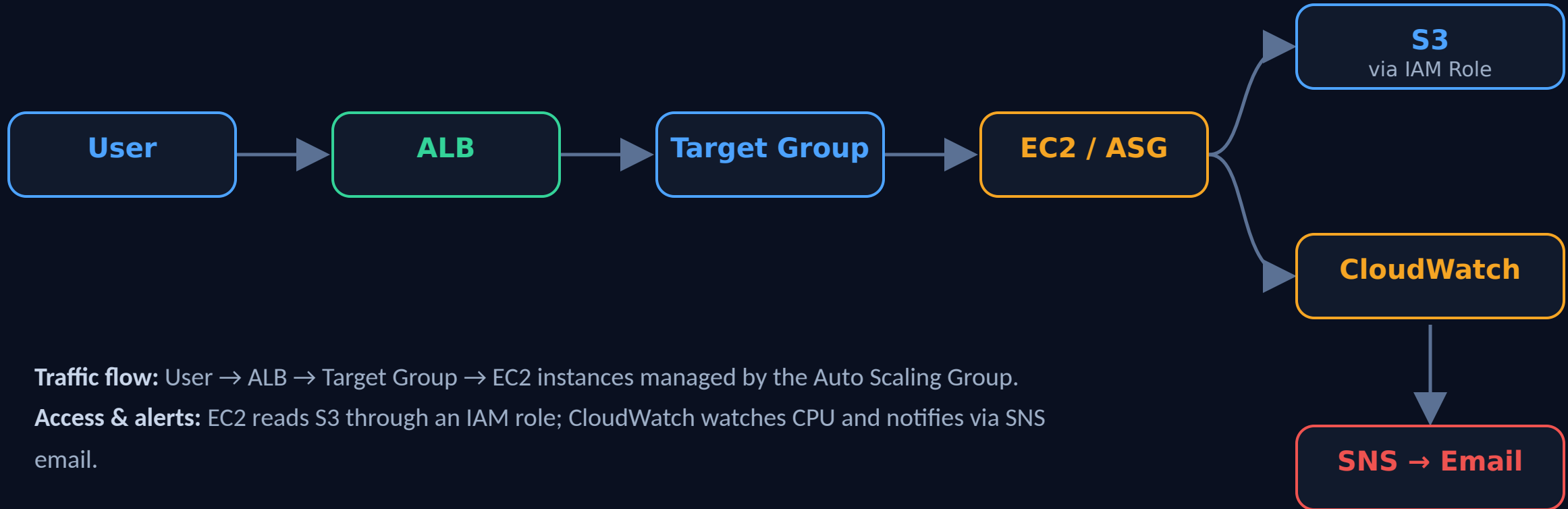
Support mindset

- Verify connectivity layer-by-layer
- Use health checks to isolate failures
- Validate IAM access from the server
- Simulate CPU pressure and observe alarm behavior
- Clean up resources after testing to control cost

2. Architecture Overview

Simplified view of the integrated lab used for testing.

VPC: 10.0.0.0/16 | Public subnets in two Availability Zones



Traffic flow: User → ALB → Target Group → EC2 instances managed by the Auto Scaling Group.

Access & alerts: EC2 reads S3 through an IAM role; CloudWatch watches CPU and notifies via SNS email.

3. VPC & Networking

Network foundation for the web tier.

10.0.0.0/16

VPC CIDR

2 AZs

ALB subnet coverage

3 subnets

2 public + 1 private

IGW

Public internet route

Networking design

- Public subnet in ap-south-1a
- Public subnet in ap-south-1b for ALB
- Private subnet reserved for internal-tier design
- Public route table used an Internet Gateway route

Troubleshooting incident

- EC2 Instance Connect initially failed
- Security group and IGW were checked
- Root cause: public subnet was not associated with the public route table
- After association, Instance Connect worked

4. Application Load Balancer

ALB provided the entry point and distributed HTTP traffic.

Configuration

- Internet-facing Application Load Balancer
- IPv4
- HTTP listener on port 80
- Mapped to public subnets in two Availability Zones
- Forwarded traffic to target group web-tg

Validation

- ALB DNS endpoint opened successfully
- Target group health checks became healthy
- Resource map confirmed listener → rule → target group flow
- ALB continued serving traffic when one target was deregistered

5. Auto Scaling Group

Capacity management and self-healing behavior.

2

Desired capacity

2 / 2

Min / Max

t3.micro

Instance type

2 AZs

Subnet distribution

Self-healing test

- One ASG-managed instance was terminated intentionally
- ASG detected capacity below desired level
- A replacement instance was launched automatically
- Desired capacity returned to 2

Support takeaway

- ASG reduces manual recovery work
- Health and capacity should be checked before blaming the load balancer
- Activity history is useful for incident investigation

6. S3 + IAM Role Integration

Secure server-side access without hard-coded AWS keys.

Access flow

- EC2 web server
- IAM instance role: ec2-s3-read-role
- Policy: AmazonS3ReadOnlyAccess
- S3 bucket with test object
- AWS CLI used from the server to verify access

CLI proof

- `aws s3 ls` → bucket visible
- `aws s3 ls s3://<bucket>` → test object visible
- `aws configure list` showed IAM role credentials
- No access key or secret key was required on the instance

7. CloudWatch + SNS Monitoring

CPU monitoring and notification flow.

Monitoring setup

- Metric: EC2 CPUUtilization
- Alarm threshold used in the lab: 60%
- CPU stress was generated on the test server
- Observed peak was above the threshold
- Alarm transitioned and notification was delivered

Incident flow

- CPU load increases
- CloudWatch evaluates the metric
- Alarm state changes
- SNS publishes notification
- Email alert reaches the subscriber

8. Troubleshooting Cases

Real incidents used to practice Cloud Support investigation.

EC2 connectivity

Symptom: Instance Connect failed

Finding / Fix: Subnet was not associated with the public route table

ALB routing

Symptom: Target state / routing checked

Finding / Fix: Listener → target group → healthy targets validated

ASG recovery

Symptom: Managed instance terminated

Finding / Fix: ASG automatically launched replacement

IAM + S3

Symptom: Server access verification

Finding / Fix: IAM role credentials successfully listed S3

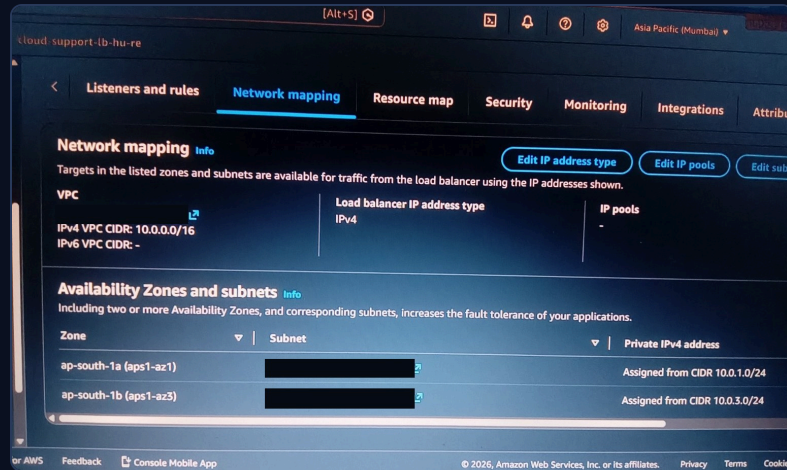
CPU alarm

Symptom: CPU exceeded threshold

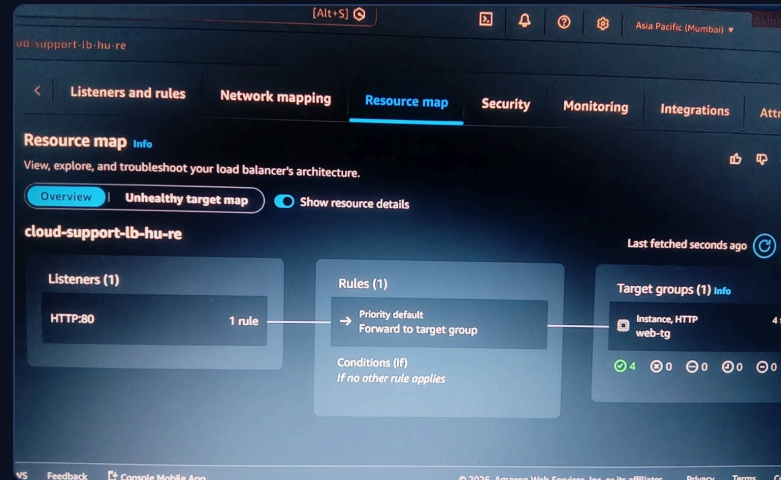
Finding / Fix: CloudWatch → SNS → email notification

9. Evidence from the AWS Console

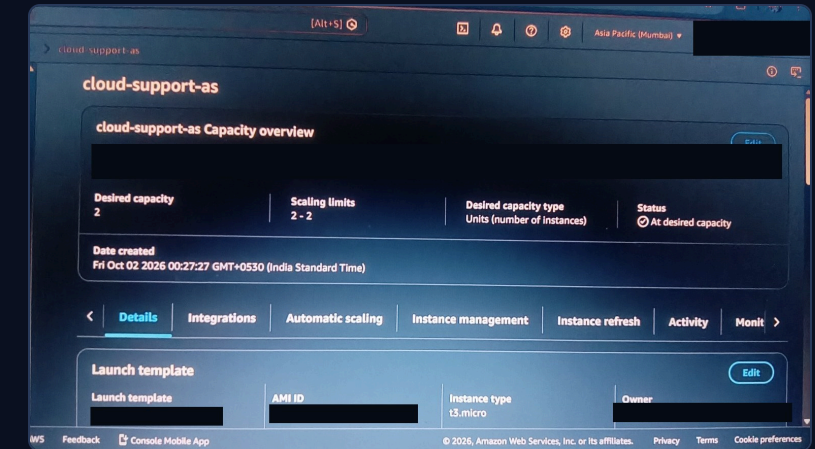
Selected screenshots from the actual lab session (account IDs and resource IDs masked).



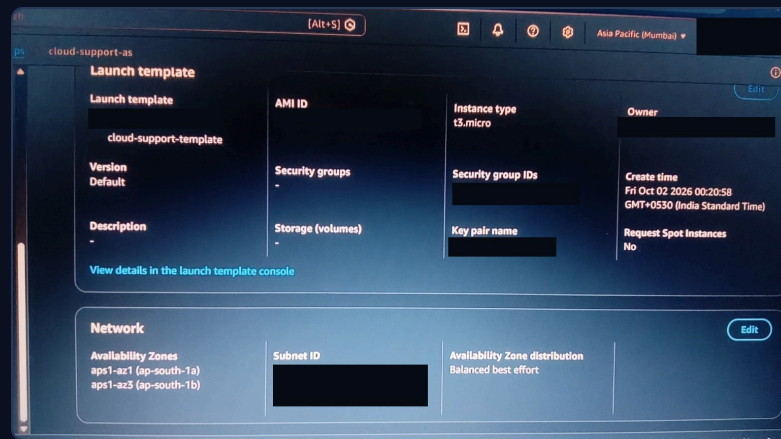
ALB network mapping



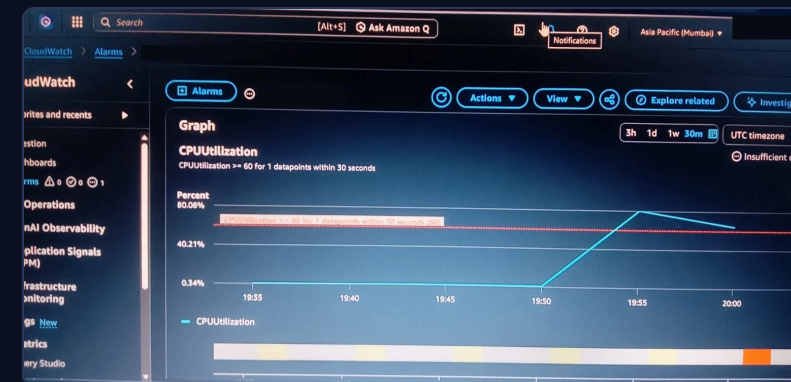
ALB resource map



ASG capacity overview



Launch template configuration



CloudWatch CPU alarm

10. Security & Cost-Control Practices

What was intentionally done to keep the lab safer and cheaper.

Security

- Used IAM role instead of hard-coded AWS access keys
- S3 public access was not required for the test
- Security groups were used to control HTTP/SSH access
- Sensitive credentials were not included in the project repository

Cleanup

- Deleted ASG, ALB, target group and EC2 instances
- Removed launch template, AMI and associated snapshot
- Deleted S3 bucket, IAM role, CloudWatch alarm and SNS topic
- Removed project VPC networking
- Final checks confirmed only default security-group resources remained

11. Project Scope & Limitations

Clear scope makes the portfolio project credible.

Implemented in this integrated lab

- VPC + public/private subnet design
- Internet Gateway + route tables
- EC2 web tier
- ALB + target group
- Auto Scaling Group
- S3 + IAM role
- CloudWatch + SNS

Not implemented in this integrated lab

- RDS
- NAT Gateway
- Public Route 53 domain mapping
- These are intentionally excluded from the integrated lab scope

12. Key Learnings

Skills demonstrated through hands-on troubleshooting.

Technical

- AWS networking and routing diagnosis
- ALB and target health troubleshooting
- Auto Scaling self-healing behavior
- IAM role-based access from EC2
- AWS CLI verification
- CloudWatch alarm and SNS notification flow

Cloud Support approach

- Start from the symptom and verify each layer
- Check configuration before changing resources
- Use logs, health checks and metrics as evidence
- Make the smallest safe fix
- Document the root cause and resolution
- Clean up after the lab